

Exhibit CC-135

TPS53840

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Enhanced high-current PMIC for DDR5 server MRDIMM

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DATA SHEET

<https://www.ti.com/lit/gpn/TPS53840> | [TPS53840](https://www.ti.com/lit/gpn/TPS53840) Integrated Step-Down Converter for DDR5 MRDIMM Server Power Supply datasheet (<https://www.ti.com/lit/gpn/TPS53840>)

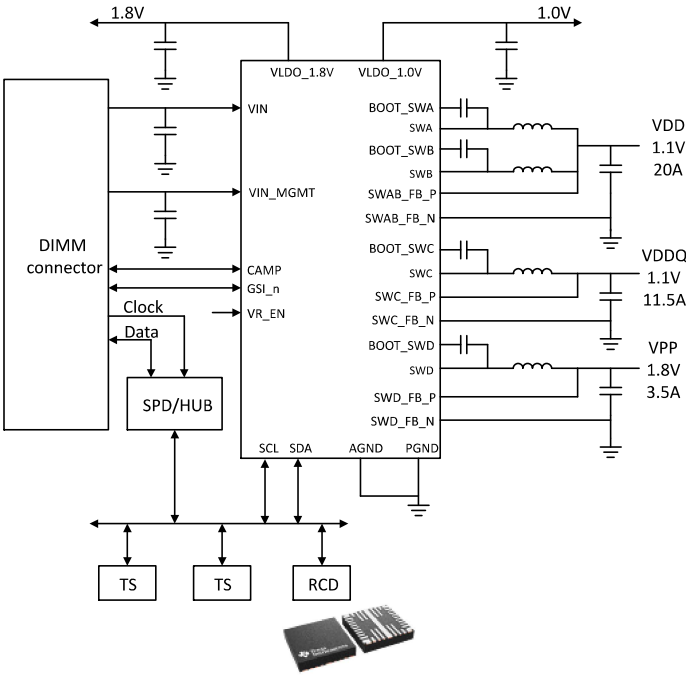
PDF (<https://www.ti.com/lit/gpn/TPS53840>) | HTML (<https://www.ti.com/document-viewer/TPS53840/datasheet>)

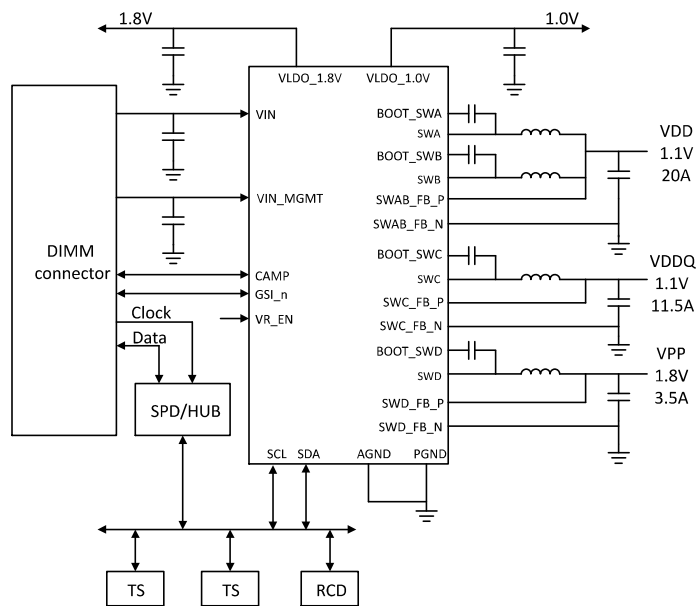
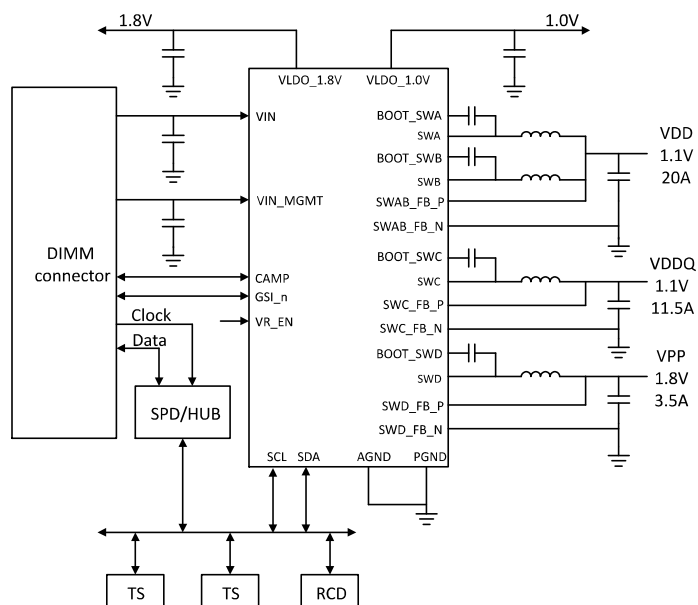
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Parameters Package | Pins | Size **Features** Description

Parameters

Package | Pins | Size

Features

- Compliant to JEDEC PMIC5020 for DDR5 MRDIMM and RDIMM
- 4 Buck regulators
 - SWA (1.1V) 8.5A DC/10A Pk
 - SWB (1.1V) 8.5A DC/10A Pk
 - SWC (1.1V) 10A DC/11.5A Pk
 - SWD (1.8V) 3A DC/3.5A Pk
- Single or dual phase for SWA and SWB

- 2 LDO: 1.8V / 25mA and 1.0V / 20mA
- VIN_Bulk (4.25V to 15V) and VIN_Mgmt (3.0V to 3.6V) input supplies
- Automatic switchover from VIN_Mgmt to VIN_Bulk
- Overvoltage, under voltage, over current, temperature warning and temperature shutdown
- Error injection capability
- DIMM specific registers for customization (EEPROM)
- Persistent error log registers
- Programmable switching frequency: 500kHz to 1.375MHz
- Power good pin (CAMP) and General status interrupt pin (GSI_n)
- I2C and I3C Bus interface for telemetry of voltage, current, power, temperature, and fault conditions
- Enable with I2C/I3C, VR_EN pin or Auto power on
- 5mm × 5mm, 35-Pin, QFN PowerPad™ package

Description

 [Find other Multi-channel ICs \(PMICs\).](/product-category/power-management/multi-channel-ics-pmic/products.html) (</product-category/power-management/multi-channel-ics-pmic/products.html>)

Request more information

The full data sheet and additional documents are available under NDA. [Request now](https://www.ti.com/drr/opn/TPS53840-DESIGN) (<https://www.ti.com/drr/opn/TPS53840-DESIGN>)

Technical documentation

Type	Title	Date 
All 	Filter title by keyword 	
★ Data sheet	(https://www.ti.com/lit/gpn/TPS53840) PDF (https://www.ti.com/lit/gpn/tps53840) HTML (https://www.ti.com/document-viewer/tps53840/datasheet)	Apr 30, 2024

Design & development

For additional terms or required resources, click any title below to view the detail page where available.

All CAD symbols, footprints & 3D models

CAD symbols, footprints & 3D models		
Package	Pins	CAD symbols, footprints & 3D models
VQFN-HR (RWZ)	35	Ultra Librarian (https://vendor.ultralibrarian.com/TI/embedded/?gpn=TPS53840&package=RWZ&pin=35&sid=019fe3cd7cae002f4cf96d109a0c0507d003007501558&c=1)

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Ordering & quality

Part number ↓↑	Buy	TI.com inventory ↓↑	Qty Price (USD) ↓↑	Package qty Carrier ↓↑	Package Pins ↓↑	Material type ↓↑	Status
				Select ▼	Select ▼		Select
✓ TPS53840RWZR Same as: TPS53840RWZR.A	Log in to order	Log in to view inventory Part notifications	1ku \$5.520 ▼	3,000 LARGE T&R CUSTOM	VQFN-HR (RWZ) 35	Production	✓ ACTIVE

Products to complete your design



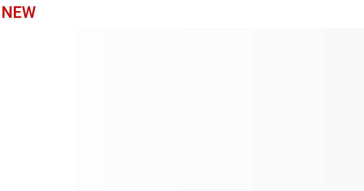
OSCILLATORS

LMK6H ✓ **ACTIVE**
Low-jitter, high-performance, bulk-acoustic-wave (BAW) fixed-frequency HCSL oscillator

Approx. price (USD) 1ku | 2.300

[Data sheet \(PDF\)](#)

[Online data sheet](#)



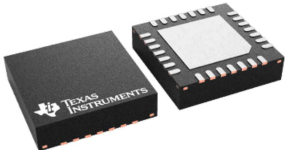
DC/DC CONVERTERS

TPS544A28 ✓ **ACTIVE**
4-V to 16-V input, 12-A, PMBus, synchronous step-down converter

Approx. price (USD) 1ku | 1.900

[Data sheet \(PDF\)](#)

[Online data sheet](#)



CLOCK BUFFERS

LMKDB1204 ✓ **ACTIVE**
2-input 4-output LP-HCSL clock multiplexer for Gen 1 to Gen 7

Approx. price (USD) 1ku | 1.860

[Data sheet \(PDF\)](#)

[Online data sheet](#)

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